there have been reports of drones being used to carry explosives, deliver biological or chemical agents, and disrupt public events or air traffic. Such misuse not only poses immediate security risks but also undermines public trust in legitimate drone applications.

To combat these threats, governments and law enforcement agencies are implementing strict registration requirements, remote identification systems, and geofencing technologies that restrict drone operation in prohibited areas. Additionally, the use of AI-based surveillance and airspace monitoring systems enables real-time detection of suspicious drone behavior. International collaboration and intelligence sharing are also critical to prevent the weaponization and cross-border misuse of drones.

7.8.2 Hacking, spoofing, and hijacking drones: Addressing cybersecurity risks

With the increasing integration of AI, GPS, and wireless communication systems, drones are vulnerable to cyberattacks that target their software, navigation, or control networks. Hacking, GPS spoofing, and signal hijacking are among the most serious cybersecurity risks, allowing attackers to manipulate or take over drones mid-flight. Such intrusions can lead to data theft, mission disruption, or physical damage, particularly in defense, surveillance, and commercial delivery applications.

GPS spoofing involves transmitting false satellite signals to mislead a drone’s navigation system, redirecting it off course or into restricted air-space. Similarly, hacking attacks may exploit weak encryption or unsecured data links to gain control of the drone’s command system. Denial-of-service (DoS) attacks can overwhelm communication channels, causing drones to crash or lose connection with operators. These threats are particularly alarming for autonomous drones relying on cloud-based data or swarm coordination systems.